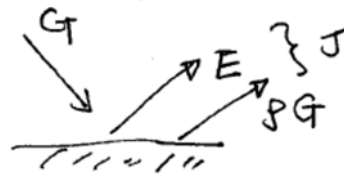
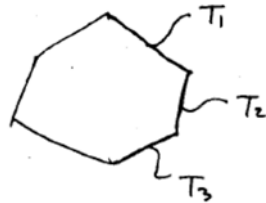


24. Radiation heat transfer between diffuse gray surfaces in an enclosure

[Assumptions]

- ♦ Isothermal surfaces
- ♦ Uniform radiosity and irradiation
- ♦ Opaque, diffuse, gray surfaces
- ♦ Non-participating (transparent) medium



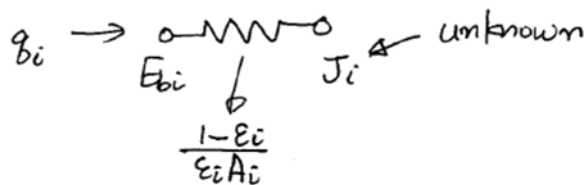
$$J = E + \rho G = E + (1 - \alpha)G = E + (1 - \varepsilon)G \quad \because \alpha = \varepsilon \text{ (Kirchhoff's law)}$$

q_i : Net radiation HT rate at which radiation leaves surface i

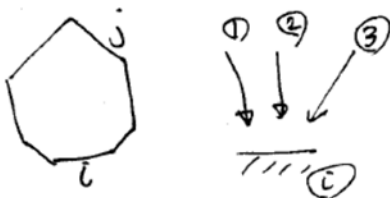


$$q_i = A_i(J_i - G_i) = A_i \left(J_i - \frac{J_i - E_i}{1 - \varepsilon_i} \right) = A_i \left(J_i - \frac{J_i - \varepsilon_i E_{bi}}{1 - \varepsilon_i} \right) \quad (*)$$

$$\therefore q_i = \frac{E_{bi} - J_i}{\left(\frac{1 - \varepsilon_i}{\varepsilon_i A_i} \right)} \Leftrightarrow i = \frac{\Delta V}{R}$$



To obtain radiosity J , irradiation G should be clarified. Note that the source of irradiation is radiosity of other surfaces



$$A_i G_i = F_{i1} A_1 J_1 + F_{i2} A_2 J_2 + F_{i3} A_3 J_3 + \dots + F_{iN} A_N J_N = \sum_{j=1}^N F_{ji} A_j J_j = \sum_{j=1}^N F_{ij} A_i J_j$$

$$\rightarrow G_i = \sum_{j=1}^N F_{ij} J_j$$

Eq. (*) yields

$$q_i = A_i(J_i - G_i) = A_i \left(J_i - \sum_{j=1}^N F_{ij} J_j \right) = A_i \left(J_i \sum_{j=1}^N F_{ij} - \sum_{j=1}^N F_{ij} J_j \right) = \sum_{j=1}^N A_i F_{ij} (J_i - J_j) = \sum_{j=1}^N \frac{J_i - J_j}{\left(\frac{1}{A_i F_{ij}} \right)}$$

Consequently,

$$q_i = \frac{E_{bi} - J_i}{\left(\frac{1 - \varepsilon_i}{\varepsilon_i A_i} \right)} = \sum_{j=1}^N \frac{J_i - J_j}{\left(\frac{1}{A_i F_{ij}} \right)} \quad (**)$$

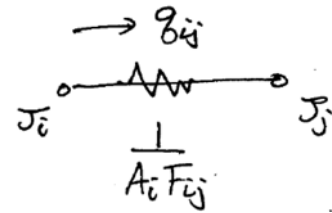
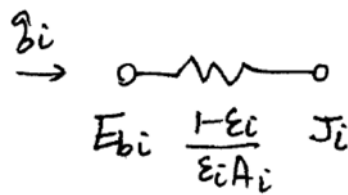
Note that $\frac{E_{bi} - J_i}{\left(\frac{1 - \varepsilon_i}{\varepsilon_i A_i} \right)} = \sum_{j=1}^N \frac{J_i - J_j}{\left(\frac{1}{A_i F_{ij}} \right)}$ give N simultaneous equations for N unknowns

$J_1, J_2, J_3, \dots, J_n$. Then, Eq.(**) gives $q_1, q_2, q_3, \dots, q_n$.

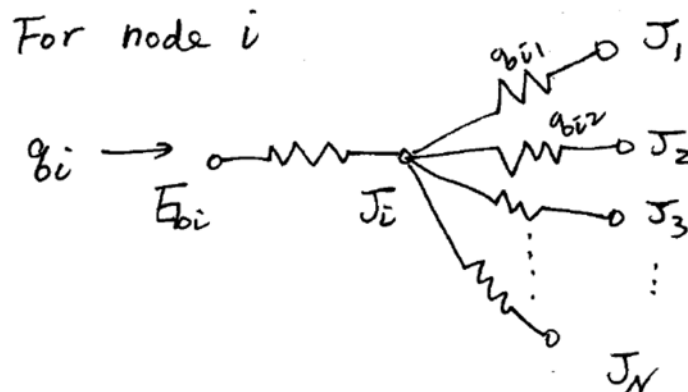
[Method using electrical circuit analogy]

Considering the surfaces as nodes, an electrical circuit (network diagram) can be formed.

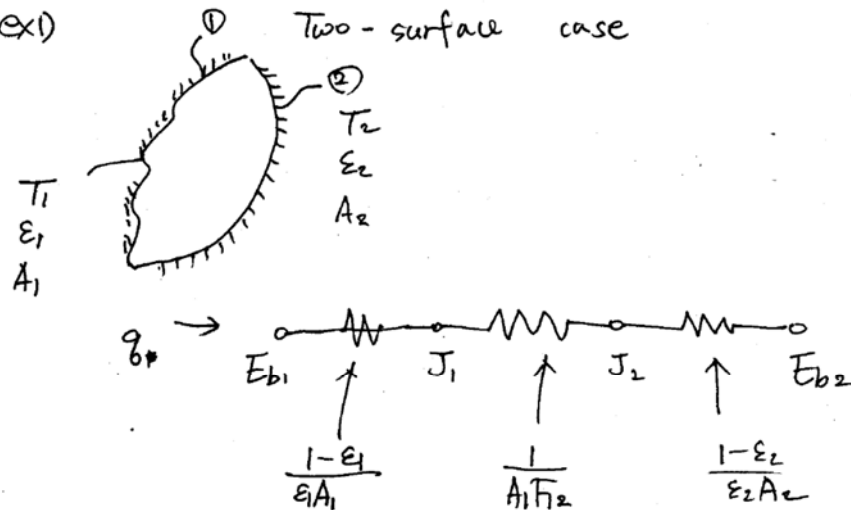
- Current $\sim$ Heat flow
- Voltage $\sim E_b$ and J
- Resistance $\sim$ Thermal resistance for radiation
 - ◆ Between E_{bi} and J_i ,
 - ◆ Between J_i and J_j ,



- For an arbitrary node (surface) i , the circuit diagram becomes



ex1)

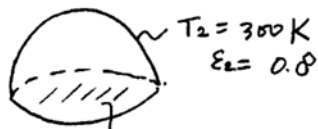


$$q_{12} = \frac{E_{b1} - E_{b2}}{R_{tot}} = \frac{E_{b1} - E_{b2}}{\frac{1-\epsilon_1}{\epsilon_1 A_1} + \frac{1}{A_1 F_{12}} + \frac{1-\epsilon_2}{\epsilon_2 A_2}} = \frac{\sigma T_1^4 - \sigma T_2^4}{\text{Oval}}$$

If $A_2 \gg A_1$ $q_{12} = \epsilon_1 A_1 \sigma (T_1^4 - T_2^4)$

(Note) The above analysis yields the Stefan Boltzmann law of radiation. The equation is valid only when the surrounding surface (A_2) is much larger than the surface of an object (A_1).

ex2)



$$T_1 = 1000K$$

$$\epsilon_1 = 0.5$$

$$A_1 = \pi \text{ m}^2$$

$$A_2 = 2\pi \text{ m}^2$$

$$R = 1\text{ m}$$

$$F_{11} = 0$$

$$F_{12} = 1$$

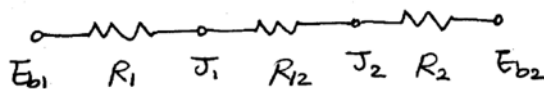
$$F_{21} = \frac{A_1}{A_2} F_{12} = \frac{1}{2}$$

$$F_{22} = \frac{1}{2}$$

$$R_1 = \frac{1-\epsilon_1}{\epsilon_1 \times \pi} = \frac{1}{\pi} \text{ m}^{-2}$$

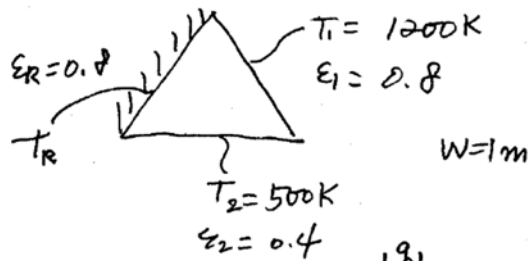
$$R_2 = \frac{1-\epsilon_2}{\epsilon_2 \times 2\pi} = \frac{1}{8\pi} \text{ m}^{-2}$$

$$R_{12} = \frac{1}{\pi \cdot \frac{1}{2}} = \frac{2}{\pi} \text{ m}^{-2}$$

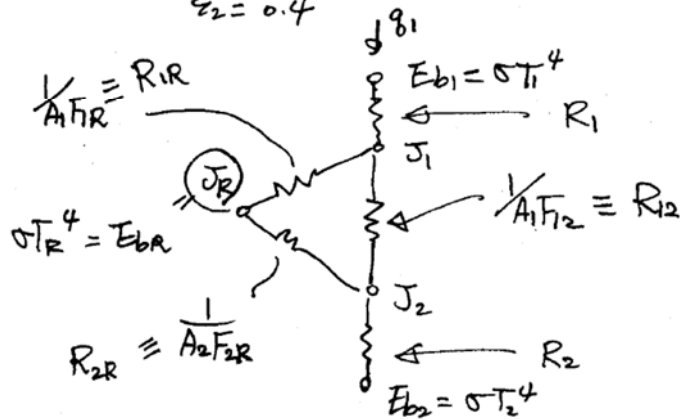


$$q_{12} = \frac{E_{b1} - E_{b2}}{R_{tot}} = \frac{\sigma (T_1^4 - T_2^4)}{R_1 + R_{12} + R_2} = \frac{\sigma (1000^4 - 300^4)}{\frac{1}{\pi} + \frac{1}{8\pi} + \frac{2}{\pi}}$$

$$= 56.5 \text{ kW}$$



$$T_R = ?$$



$$F_2 = F_R = F_{R2} = \frac{1}{2}$$

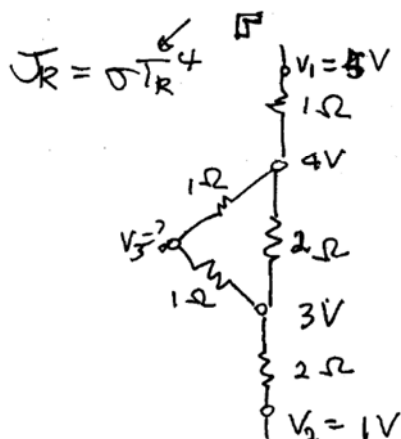
$$q_1 = \frac{E_{b1} - E_{b2}}{R_1 + R_2 + \left\{ \frac{1}{R_{12}} + \frac{1}{R_{1R} + R_{2R}} \right\}^{-1}}$$

$$\frac{E_{b1} - J_1}{R_1} = q_1$$

$$\frac{J_2 - E_{b2}}{R_2} = q_1$$

$$\frac{J_1 - J_R}{R_{1R}} = \frac{J_R - J_2}{R_{2R}}$$

$$\text{or } \frac{J_1 - J_2}{R_{1R} + R_{2R}} = \frac{J_1 - J_R}{R_{1R}} \dots$$



$$R_{tot} = 1 + 2 + \frac{1}{\frac{1}{2} + \frac{1}{2}} = 4\Omega$$

$$i_1 = \frac{V_1 - V_2}{R_{tot}} = \frac{5 - 1}{4} = 1A$$

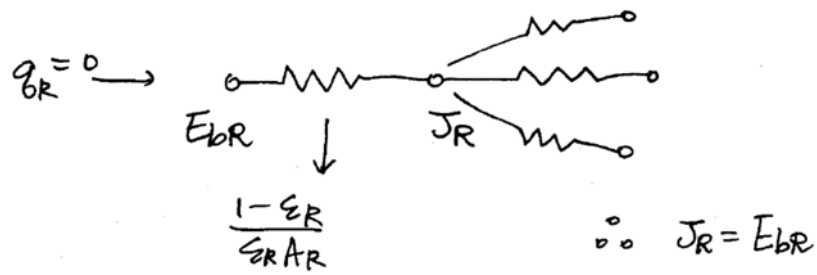
$$\frac{4 - V_3}{1} = \frac{V_3 - 3}{1} \quad 2V_3 = 7$$

$$V_3 = 3.5V$$

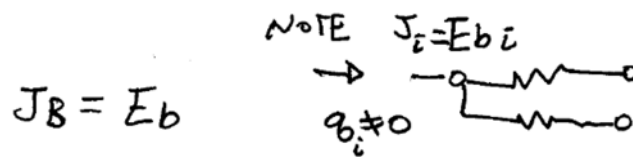
□

[Special surfaces]

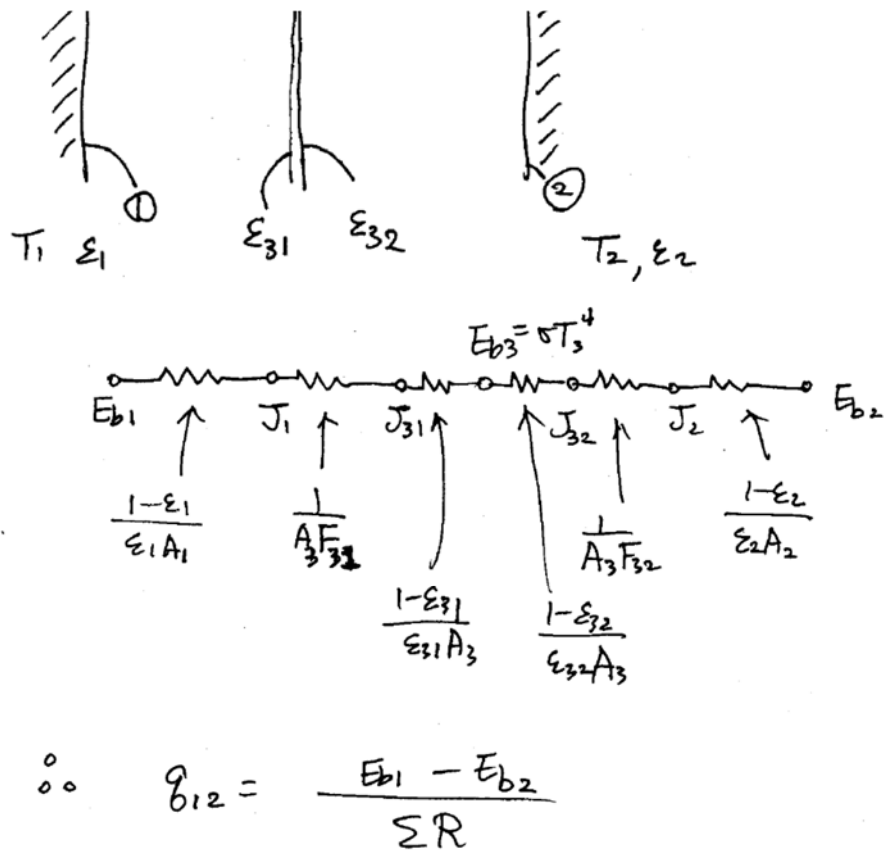
- Reradiating (adiabatic, insulated) surface



- Black surface



- Radiation shield (plate made of high-reflectivity, i.e., low-emissivity, material)



<Summary - review of radiation network scheme>

$$q_i = \frac{E_{bi} - J_i}{\left(\frac{1 - \varepsilon_i}{\varepsilon_i A_i} \right)} = \sum_{j=1}^N \frac{J_i - J_j}{\left(\frac{1}{A_i F_{ij}} \right)}$$

- ① Draw all nodes corresponding to the surfaces (Label them as J_i 's)
- ② For each radiosity nodes (J_i 's), draw blackbody emissive power nodes (E_{bi} 's)
- ③ Calculate all F_{ij} 's
- ④ Compute resistances
 - $\frac{1 - \varepsilon_i}{\varepsilon_i A_i}$ between E_{bi} and J_i
 - $\frac{1}{A_i F_{ij}}$ between J_i and J_j ,
- ⑤ Regard J_i 's and E_{bi} 's as voltage and calculate electric current i 's

Current $i \Leftrightarrow$ Net radiation heat transfer between the corresponding surfaces